

ANT-EV

Focused station charging allocation guide

ANT-EV is a station-level demonstration of how limited charging power is split across vehicles. The product intentionally contains only six areas: Charge Split, Models, Emergency Response, Architecture, Patent Specification, and Help. It is a simulation, not a live charger controller.

1. What the station does

The station receives vehicle-session and station-state inputs, predicts the conditions that matter to charging, removes unsafe choices, and distributes the site power limit between eligible vehicles. Every shown allocation includes a human-readable reason.

Step	Station action
1. Read	Read session urgency, requested readiness, battery and station state.
2. Predict	M1-M5 generate decision-support signals.
3. Score	PRISM-ANT creates a priority score from urgency, deadline, flexibility, forecast pressure and battery impact.
4. Protect	Safety Guard rejects charger, transformer, vehicle and battery-limit violations.
5. Split	M6 ranks feasible strategies; capped priority water-filling shares the available site power.
6. Explain	Show allocated kW, reason, and safety outcome for every vehicle.

2. Charging split algorithm - patent demonstration

Name: Safety-Gated Priority Capped Water-Filling (SG-PCW).

For each vehicle i , ANT-EV calculates a priority score from PRISM-ANT, applies deterministic safety checks, and limits the vehicle to its safe charging cap. Remaining station power is distributed proportionally to safe priority until a vehicle reaches its cap, then the remaining power is redistributed among the other eligible vehicles.

Formula: $P_i = \text{safe}(\text{priority}_i) \times \text{cap}_i$; $p_i = \min(\text{cap}_i, \text{remaining} \times P_i / \sum(P))$.

The allocator fills all available station capacity whenever active sessions can safely accept it. If every remaining session has reached its safe cap, unused power is intentionally left unused. Emergency priority can raise a session score but cannot override safety caps.

3. Model input/output specification

Model	Understandable input	Output	How it helps the split
M1 Energy need	session arrival, historic session context	energy need and charging duration	Avoids assigning more power than a session needs.
M2 Station demand	time, queue, capacity, current station load	future station load	Keeps headroom when demand is predicted to rise.
M3 Battery condition	compatible cycle, capacity, resistance data	SOH/RUL or Limited Battery Data	Prevents unsupported health claims.
M4 Charging stress	temperature, C-rate, SOC change	stress score and safe power envelope	Reduces a vehicle cap when fast charging is stressful.
M5 Future state	ordered SOC, temperature, power, grid sequence	future trajectory	Checks whether the current split remains safe.
M6 Strategy ranking	safe candidates and M1-M5 outputs	ranked allocation strategy	Selects a feasible station-wide split.

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Emergency, architecture, demos and operation

4. Emergency response

Scenario	Station reaction
Medical emergency	Verify emergency context and required readiness; run Safety Guard; give the safe emergency session a higher priority score; reduce only flexible lower-priority allocations; record the reason.
Goods-delivery emergency	Verify the time-critical goods requirement; run the same Safety Guard; raise priority only within the site limit; preserve required medical priority if both occur.

Neither medical nor goods priority bypasses the charger limit, site transformer limit, vehicle maximum charge power, or battery-safe envelope.

5. Architecture

Vehicle + station inputs -> M1-M5 predictions -> PRISM-ANT priority -> Safety Guard -> M6 strategy ranking -> station-safe power split -> allocation reason. Vehicles do not negotiate with one another; the station is the only decision authority.

6. Included demonstrations

Demo	Purpose
10 Vehicle Station	Ten simultaneous vehicle sessions are evaluated and split under one 300 kW station limit.
Emergency Commute	A medical emergency and goods-delivery urgency are evaluated using the same safety-constrained allocation method.

7. Run the project

Web application

1. npm install
 2. npm run dev
 3. Open the local URL printed by Vite, normally `http://localhost:5173`
 4. Select a demo in the top bar, optionally add or remove an EV, then press Run demo or the step button.
- Completion immediately redistributes released power.

Training

The currently data-backed M1 and M2 baseline trainers run with: `python training/export_models.py`. M3-M6 must not be represented as trained until their data-backed trainer pipelines are implemented and validated.

Scope and safety notice

This PDF documents a local demonstration. It must not be used to control charging hardware without validated models, approved station configuration, authenticated telemetry, and a production-grade deterministic safety layer.